



Time = 20 Minutes

INSTRUCTOR SOLUTION - Quiz No 04

[20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.

CLO Assessed – Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3) – (applicable to both the questions.)

Question 1 [10 points] [Probability Mass Function of a Random Variable]

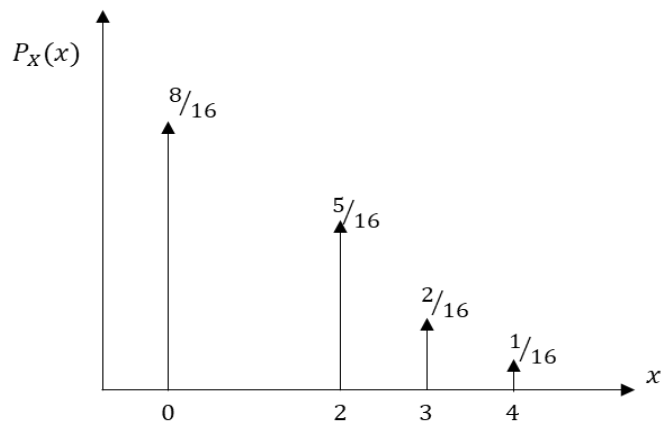
We toss a fair coin 4 times. We define a Discrete Random Variable X as the number of consecutive Heads (we count a minimum two consecutive Heads) in 4 throws of the coin. Therefore, $X = 0$ if the outcome of the 4-throw sequence is H, T, H, T or T, T, H, T and $X = 3$ if the sequence is T, H, H, H.

Part a [5 points] Find and plot the probability mass function p.m.f for the DRV X .

Part b [5 points] Find the mean and variance of X .

(a)

Possible Outcomes				R.V X	Probabilities
T	T	T	T	0	$1/16$
T	T	T	H	0	$1/16$
T	T	H	T	0	$1/16$
T	T	H	H	2	$1/16$
T	H	T	T	0	$1/16$
T	H	T	H	0	$1/16$
T	H	H	T	2	$1/16$
T	H	H	H	3	$1/16$
H	T	T	T	0	$1/16$
H	T	T	H	0	$1/16$
H	T	H	T	0	$1/16$
H	T	H	H	2	$1/16$
H	H	T	T	2	$1/16$
H	H	T	H	2	$1/16$
H	H	H	T	3	$1/16$
H	H	H	H	4	$1/16$



(b)
$$E(X) = 2 \cdot \frac{5}{16} + 3 \cdot \frac{2}{16} + 4 \cdot \frac{1}{16} = \frac{10}{8} = 1.25$$

$$E(X^2) = 2^2 \cdot \frac{5}{16} + 3^2 \cdot \frac{2}{16} + 4^2 \cdot \frac{1}{16} = \frac{54}{16} = 3.375$$

$$Var(X) = E(X^2) - E(X)^2 \quad (\text{we studied this formula for Variance in Lecture No 13})$$

$$Var(X) = 3.375 - 1.25^2$$

$$= 3.375 - 1.5625 = 1.8125$$

Question 2 [10 points] [Probability Mass Function of a Random Variable]

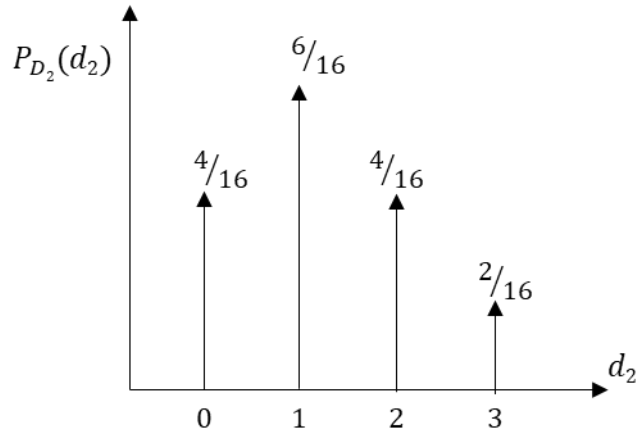
We have a fair tetrahedral die with numbers 1, 2, 3 and 4. We roll the tetrahedral die n times and note the minimum value, m_n and the maximum value, M_n appearing during the n rolls. We define the Random Variable $D_n = M_n - m_n$ as the difference between the maximum and minimum values appearing in n rolls. As an example, for a three rolls sequence, if a particular outcome is 1, 3, 1, then the R.V, $D_3 = 2$ for that particular sequence.

Part a [6 points] Find and plot the probability mass function p.m.f for the DRV D_2 . (Note - this is for two throws of a tetrahedral die.)

Part b [4 points] Just based on intuition, how do you think the distribution of Random Variable D_{16} behave?

(a)

Possible Outcomes		R.V D_2	Probabilities
roll 1	roll 2		
1	1	0	$1/16$
1	2	1	$1/16$
1	3	2	$1/16$
1	4	3	$1/16$
2	1	1	$1/16$
2	2	0	$1/16$
2	3	1	$1/16$
2	4	2	$1/16$
3	1	2	$1/16$
3	2	1	$1/16$
3	3	0	$1/16$
3	4	1	$1/16$
4	1	3	$1/16$
4	2	2	$1/16$
4	3	1	$1/16$
4	4	0	$1/16$



(b)

When we roll the tetrahedral die many times (16 times as in D_{16}), then there is sufficient probability of getting a number 4 and number 1 amongst the 16 rolls. Therefore, the p.m.f for the R.V D_{16} , will have very high probability for value 3 (approximately ≈ 1). It is very unlikely that amongst the 16 rolls you don't get 1 & 4.

The probability for 0 will be extremely low, all 16 numbers being same – this probability is $4 \cdot (1/4)^{16}$ (this is approximately ≈ 0). Similarly, to get the difference of 1 or 2, is very, very low as there is most likely a 1 and a 4 in the 16 rolls.

